6.033 Spring 2018

Lecture #14

- **Reliability via Replication**
  - **General approach to building fault-tolerance systems**
  - **Single-disk failures: RAID**

How to Design Fault-tolerant Systems in Three Easy Steps

1. identify all possible faults

Windows

A fatal exception 0E has ocurred at 0028:C0011E36 in VXD VMM(01) +
00010E36. The current application will be terminated.

* Press any key to terminate the current application.
* Press CTRL+ALT+DEL again to restart your computer. You will
lose any unsaved information in all applications.

Press any key to continue _

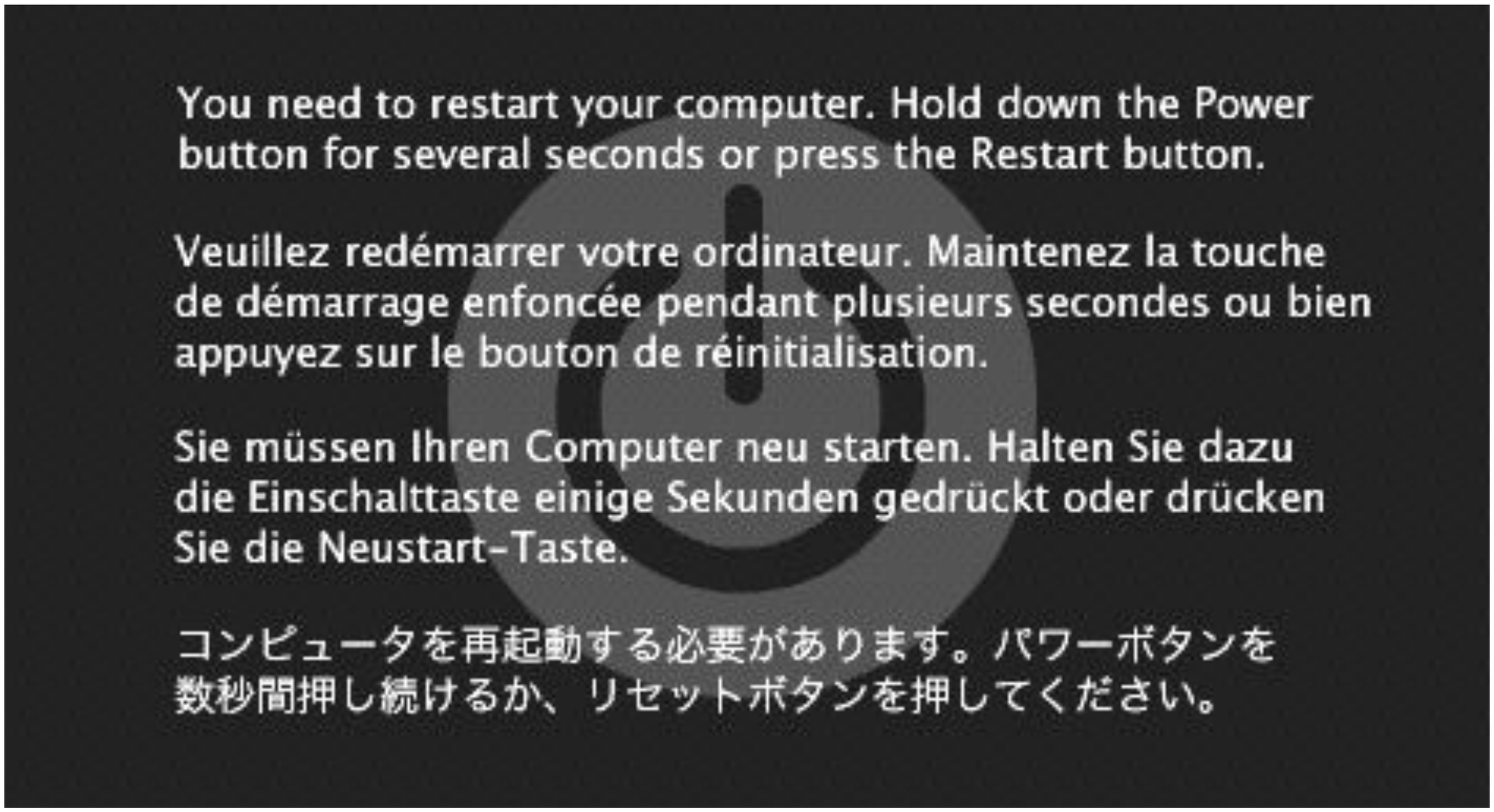
:(

Your PC ran into a problem and needs to restart. We're just collecting some error info, and then we'll restart for you.

25% complete

For more information about this issue and possible fixes, visit http://windows.com/stopcode

If you call a support person, give them this info:
Stop code: CRITICAL_PROCESS_DIED


You need to restart your computer. Hold down the Power button for several seconds or press the Restart button.
Veuillez redémarrer votre ordinateur. Maintenez la touche de démarrage enfoncée pendant plusieurs secondes ou bien appuyez sur le bouton de réinitialisation.
Sie müssen Ihren Computer neu starten. Halten Sie dazu die Einschalttaste einige Sekunden gedrückt oder drücken Sie die Neustart-Taste.
コンピュータを再起動する必要があります。パワーボタンを数秒間押し続けるか、リセットボタンを押してください。

How to Design Fault-tolerant Systems in Three Easy Steps

1. identify all possible faults
2. detect and contain the faults
3. handle the fault

quantifying reliability

dealing with disk failures

Public adaptation / original page 9

Original vendor image omitted: it is explicitly excluded from the OCW Creative Commons license.

Teaching point retained: a quoted MTBF is not an individual lifetime guarantee.

700,000 hours / (24 x 365) = 79.91 years

Illustrative aggregate estimate from the official lecture outline:

1,000 disks x 3,000 hours / 4 failures = 750,000 device-hours per failure

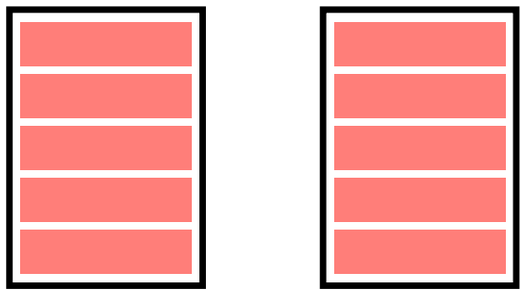
The aggregate example is hypothetical; it is not a claim about the vendor's actual test.

View the original image in the official PDF linked in examples/source.json.

Source lecture: Katrina LaCurts / MIT 6.033 / Spring 2018 / original p.9

dealing with disk failures

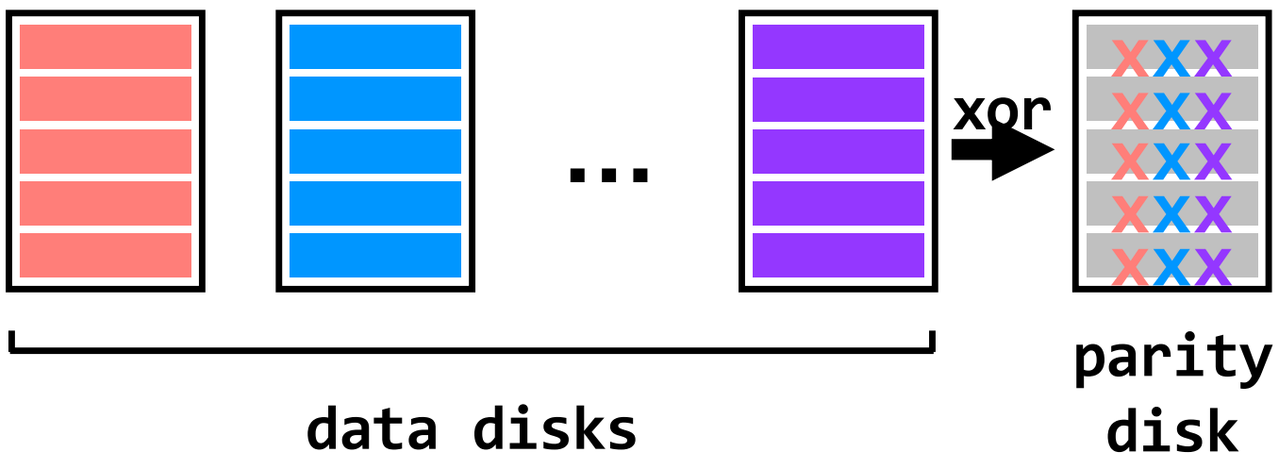
RAID 1 (mirroring)



😀 can recover from single-disk failure

😭 requires 2N disks

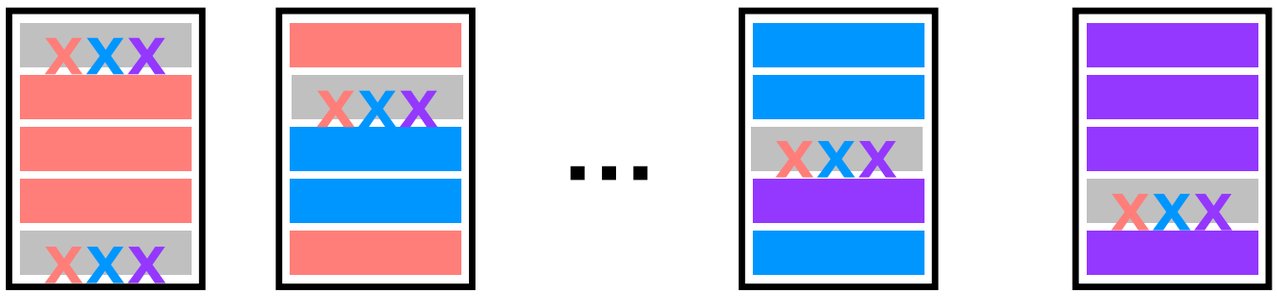
RAID 4 (dedicated parity disk)



sector i of the parity disk
is the xor of sector i
from all data disks

- 😀 can recover from single-disk failure
- 😀 requires N+1 disks (not 2N)
- 😀 performance benefits if you stripe a single file across multiple data disks
- 😭 all writes hit the parity disk

RAID 5 (spread out the parity)



😀 can recover from single-disk failure

😀 requires N+1 disks (not 2N)

😀 performance benefits if you stripe a single file across multiple data disks

😀 writes are spread across disks

- Systems have faults. We have to take them into account and build reliable, **fault-tolerant systems**. Reliability always comes at a cost — there are tradeoffs between reliability and monetary cost, reliability and simplicity, etc.

- Our main tool for improving reliability is **redundancy**. One form of redundancy is **replication**, which can be used to combat many things including disk failures (important, because disk failures mean lost data).

- **RAID** replicates data across disks in a smart way: RAID 5 protects against single-disk failures while maintaining good performance.

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